



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Energy 2 — Kinetic Energy (Homework)

This is the only physics you will do before I next see you, so give it a proper go. Show every step: an answer with no working earns almost nothing in the exam, and I cannot see where you went wrong.

1) **State** the kinetic energy equation from memory, *without looking it up*, and give the unit of each quantity. [3]

2) **Calculate** the energy in the kinetic store of a 60 kg runner moving at 5.0 m/s. [3]

3) **Calculate** the energy in the kinetic store of a 7500 kg lorry moving at 12 m/s. [3]

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4) **Compare** a 1000 kg car at 20 m/s with a 4000 kg lorry at 10 m/s. Which has the larger kinetic store? Support your answer with a calculation. [4]

5) **Determine** the speed of a 0.15 kg cricket ball whose kinetic store is 120 J. [4]

Rearranging: You are given E_k and m and want v . Rearrange the equation *before* you put any numbers in — it is easier and it earns marks.

6) **Explain** why a car travelling at 60 mph is far more than twice as dangerous as the same car at 30 mph. Refer to the kinetic store in your answer. [4]

Total: 21 marks. Bring this to the next lesson. If you are stuck, the course page has a walkthrough: rawsonvault.com/physics/aqa-gcse/p1-energy